Extended reading — Week 4: TD, SARSA, Q-learning, Expected SARSA

STUDY-RL

November 8, 2026

Read with a question in mind

Start with the week's teaching guide and its small worked example. Then use these readings to connect the algorithm to its assumptions and evidence. The explanations and investigations below are course commentary, not quotations. Read the source before reusing a theorem or a reported result. For each paper, record the objective, observation model, claimed result, conditions and one condition your local experiment does not reproduce.

1. Sutton (1988): Learning to Predict by the Methods of Temporal Differences

Primary source or book record: https://doi.org/10.1007/BF00115009.

What to learn. Introduces a family of temporal-difference prediction methods that learn from changes between successive predictions.

What the result requires. Bootstrapping trades Monte Carlo outcome noise against value approximation error. It is not an unconditional claim of unbiasedness or superiority over Monte Carlo.

Notebook investigation. Use `exercise_1_td0`: reward 1, discount 0.9, next value 2 and current value 0.5 give TD error 2.3. At learning rate 0.1 the new value is 0.73.

2. Watkins and Dayan (1992): Q-learning

Primary source or book record: https://doi.org/10.1007/BF00992698.

What to learn. Provides a convergence result for tabular Q-learning under conditions including adequate repeated sampling and appropriate learning rates.

What the result requires. The theorem does not cover a finite replay buffer and an arbitrary deep network. Discounted bounded-reward assumptions and the correct treatment of terminal states are part of the result.

Notebook investigation. Use `exercise_4_qlearn_step`. Compare a terminated transition with a time limit on a continuing task; only the former has necessarily zero continuation value.

3. van Hasselt (2010): Double Q-learning

Primary source or book record: https://proceedings.neurips.cc/paper/2010/hash/091d584fced301b442654dd8c23b3fc9-Abstract.html.

What to learn. Decouples action selection and action evaluation to address maximisation bias from noisy estimates.

What the result requires. Double estimators can have residual bias, including underestimation. The minimum of two critics in TD3 is a different construction from selecting with one estimate and evaluating with the other.

Notebook investigation. Use `exercise_6_double_q_step`. With table A preferring action 0 but table B assigning action 1 a larger value, select with A and read B at that selected action.

4. van Seijen et al. (2009): A Theoretical and Empirical Analysis of Expected Sarsa

Primary source or book record: https://doi.org/10.1109/ADPRL.2009.4927542.

What to learn. Averages the next-action value under the behaviour policy instead of sampling the next action.

What the result requires. Conditioning on the same next state and fixed value estimates removes next-action sampling variance; equality is possible when action values are equal. This does not establish uniformly lower total training error.

Notebook investigation. In `exercise_5_expected_sarsa_target`, use next values $(2, 4)$ and probabilities $(0.25, 0.75)$: the expected continuation is 3.5. The transition and reward may remain random.

5. Sutton and Barto (2018): Reinforcement Learning, Chapters 6–7

Primary source or book record: http://incompleteideas.net/book/the-book-2nd.html.

What to learn. Connects TD prediction, SARSA, Q-learning and multi-step methods through explicit backup diagrams.

What the result requires. An on-policy SARSA target follows the sampled behaviour action; Q-learning uses a greedy target while potentially exploring in behaviour. Convergence requires more than merely reducing epsilon.

Notebook investigation. Draw the sampled and averaged branches for all three targets before running the cliff-walking bake-off. Plot evaluation return separately from the cost of exploratory behaviour.

Synthesis: evidence you should be able to show

1. Reproduce one numerical calculation before running a long experiment.
2. Name the precise objective, sampling distribution and terminal convention.
3. Compare the local implementation with one source: list a shared mechanism and one simplification that prevents calling it a full reproduction.
4. Separate an algebraic identity, an assumption-dependent theorem and an empirical result. A plot alone does not prove a theorem.
5. Save the notebook with seed, parameters, axes, denominators and a short explanation of an unexpected result. Keep evaluation data separate from tuning.

Return to the teaching guide's notebook investigation for the local function names, expected checks and interpretation questions. Source code and notebooks are available together in the course repository; the web reader is the entry point and the downloaded notebook is the execution workspace.

Statistics reading bridge

Conditional averaging removes one specific source of noise. Use conditional expectation and variance to identify exactly what Expected SARSA averages. Verify the 6.75 continuation variance before comparing target variances; do not transfer that limited calculation into an unconditional performance guarantee.

Checked textbook locations: Siegrist § § 4.3, 4.7, PDF pp.460, 507. https://stats.libretexts.org/@go/page/10163. These locations support the probability and estimation concepts; the algorithm-specific results remain attributed to the primary readings above. The course source catalog distinguishes the original textbook from its supplied instructor manual and records PDF-viewer pagination.